Zihan Teng

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LLM inference acceleration across algorithms, systems, and CUDA kernels. Experience in KV cache compression, low-bit quantization, and 10,000-GPU-scale data production. First author of **SlimKV, EMNLP 2026 Main Conference Oral**.

EDUCATION

University of Science and Technology of China Sep. 2024 - Jun. 2027 (Expected)

M.S., Electronic Science and Technology · School of Microelectronics

GPA 3.91/4.3 · Rank 2/23. Visiting student, NTU, Singapore (Aug. 2025 - Jun. 2026).
Graduate Academic Scholarships (First & Second Class); ChipON Scholarship.

University of Science and Technology of China Sep. 2020 - Jun. 2024

B.S., Microelectronics and Solid-State Electronics · School of Physical Sciences

GPA 3.58/4.3 · Rank 8/39. Exchange, National Tsing Hua University, Taiwan (Jul. - Aug. 2023).
Outstanding Student Scholarships (Bronze, twice); CGN Scholarship.

INTERNSHIP EXPERIENCE

Tencent (TEG) Jun. 2026 - Present

AI Infrastructure Intern (Qingyun Program) · Data Computing Platform Department · Beijing

- **CUDA attention.** Migrated per-token-head FP8 attention from Triton to FlashAttention-3/CUTLASS for vLLM. Paged-KV addressing deduplication, warp-cooperative loads, and scale prefetching cut **kernel latency by 47%** and global loads by **88.8%**. K-centering and QK equalization enabled **1.85× prefill speedup** over BF16. Built a PoPE FA3 backend with trigonometric lookup and matmul fusion. Deployed at 10,000-GPU scale; preparing an MLSys 2027 submission.
- **Multimodal inference.** Used Nsight Systems/Compute to guide operator refactoring and fusion, pipeline parallelism, multi-worker scheduling, dynamic batching, and FP8 KV caching. Achieved **9.46× end-to-end Text-to-Sound throughput** and **16.5% higher Audio-in-Video throughput**.
- **AI numerical debugging.** Built a skill across PyTorch, vLLM, and Torch Compile for cross-architecture semantic alignment, tensor comparison, and kernel root-cause analysis. Validated on Qwen3-VL and other production workloads; reduced manual debugging from weeks to hours.

Floating AI Jun. 2024 - Aug. 2024

Machine Learning Algorithm Intern · R&D Department · Shanghai

- Developed two-stage airport-apron calibration with global parameters and polynomial surface fitting: **0.1 m-level far-field accuracy** and **1-3 minute calibration**. Integrated YOLO real-time detection with error below 10% for collision warnings.

RESEARCH EXPERIENCE

SlimKV: Joint Token-Feature KV Cache Compression Mar. 2026 - Jun. 2026

First author · EMNLP 2026 Main Conference, Oral · arxiv.org/abs/2610.02953

- Proposed query-independent token-feature compression using beacon aggregation, low-rank constrained training, SVD initialization, and layer-adaptive ranks. Removed RoPE only from beacon keys, preserving implicit positions and enabling **reconstruction-free latent attention** via matmul associativity.
- On Llama-3.1-8B and Qwen2.5-14B, exceeded LongBench baselines at **16×-32× compression** and retained **over 96%** of full-precision performance at 4×-8×. At 128K context: **7.34× attention** and **3.38× end-to-end decoding speedups**.

Cross-Layer KV Cache Compression and Inference Acceleration Jun. 2025 - Oct. 2025

- Replaced KV caches with low-rank hidden states via cross-layer SVD. On Llama-3.1-8B: **4× compression**, **0.99 cosine similarity**, and a **0.13-point LongBench drop**. CKA-guided offline U precomputation and online V mapping cut per-layer SVD from **19 s to 1.5 s (12.67×)** at 64K; compression took under 30% of prefill time.

BitsMoE: Low-Bit Mixture-of-Experts Quantization Aug. 2025 - Jan. 2026

Second author · ICLR 2027 submission · arxiv.org/abs/2606.00079

- Contributed to shared spectral decomposition (SSD) and a quantization cost model (FQCM) using spectral importance, activation sensitivity, and distortion. Refined the ILP objective and cross-layer bit allocation under memory budgets; reproduced baselines and evaluated multiple MoE models.
- Qwen3-30B-A3B at 2 bits: **64.29% accuracy across seven tasks**, exceeding MoEQuant/GEMQ by **5.96/2.80 points**; **16.47× faster** offline quantization than GEMQ.

SKILLS

Tools: Python, C, Shell, Linux, Git, LaTeX. **GPU & ML:** CUDA, PyTorch, Transformers, LoRA, vLLM, FlashAttention, Nsight Systems/Compute.
Focus: Long-context inference, KV cache and prompt compression, MoE quantization, AI agents, kernel optimization.
English: CET-6; research reading, presentations, and basic academic communication.